

Majid Ghasemi

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Research interests

Social Reinforcement Learning, Multi-agent Reinforcement Learning, Machine Ethics, AI Alignment, and AI Ethics

Education

University of Waterloo

Waterloo, ON, Canada

Ph.D. in Electrical & Computer Engineering

Sep 2025 – Present

Supervised by Mark Crowley

Research area: *Ethical decision making in multi-agent systems and how social learning can be a way to achieve it*

Wilfrid Laurier University (WLU)

Waterloo, ON, Canada

MASc in Applied Computing

Sep 2023 – May 2025

Supervised by Dariush Ebrahimi. *GPA: A+.*

Thesis title: *Reinforcement Learning for Scheduling and Routing: Electric Vehicles Charging and Patrol Scheduling*

Imam Khomeini Intl. University (IKIU)

Qazvin, Iran

B.Sc. in Software Engineering

Sep 2017 – March 2022

Supervised by Mahdi Bahaghighat. *GPA: A-.*

Publications

Learning When to Trust in Contextual Social Bandits

Majid Ghasemi, and Mark Crowley.

19th European Workshop on Reinforcement Learning

Real-Time Electric Vehicle Routing and Charging Optimization with Battery Life Consideration

Dariush Ebrahimi, Majid Ghasemi, Mohammad Kashefi, and Fadi Alzhouri.

Internet of Things journal

Rethinking AI Alignment: From Static Rewards to Social Reinforcement Learning

Majid Ghasemi, and Mark Crowley.

ICML 2026 Pluralistic ALIGNment Workshop

Can Standard MARL Metrics Distinguish Communicative from Strategic Action?

Majid Ghasemi, and Mark Crowley.

ICML 2026 PhilML Workshop

Citation count prediction based on Google Scholar profiles and Clarivate's journal citation reports

Mahdi Bahaghighat, Leila Akbari, *Majid Ghasemi*, and Qin Xin.
Information Research journal

Toward Virtuous Reinforcement Learning: A Critique and Roadmap

Majid Ghasemi, and Mark Crowley.
Machine Ethics: from formal methods to emergent machine ethics workshop at the AAAI conference

Adaptive and AI-Driven Vehicle Patrol Scheduling with Integrated Emergency Response

Majid Ghasemi, and Dariush Ebrahimi.
IECON 2025 – 51st Annual Conference of the IEEE Industrial Electronics Society.

Optimal and Heuristic Solutions for Efficient Vehicle Patrol Scheduling in Dynamic Urban Safety

Majid Ghasemi, Ebrahim Sarkhouh, Fadi Alzhouri, and Dariush Ebrahimi.
2025 IEEE 34th International Symposium on Industrial Electronics (ISIE).

Comparative Analysis of Reinforcement Learning for Reliable Real-Time EV Routing and Charging

Majid Ghasemi, Fadi Alzhouri, and Dariush Ebrahimi.
2025 21st International Conference on the Design of Reliable Communication Networks (DRCN).

Comprehensive Survey of Reinforcement Learning: From Algorithms to Practical Challenges

Majid Ghasemi, Amirhossein Mousavi, and Dariush Ebrahimi.
Preprint on Arxiv.

Introduction to Reinforcement Learning

Majid Ghasemi, and Dariush Ebrahimi.
Preprint on Arxiv.

Navigating Smart Order Routing in Fixed Income Trading with Linear Programming and Dijkstra's Algorithm

Cassel Robson, Marko Misic, Erika Capper, Mila Cvetanovska, *Majid Ghasemi*, Fadi Alzhouri, and Dariush Ebrahimi.
2024 International Conference on the Dynamics of Information Systems.

A high-accuracy phishing website detection method based on machine learning

Mahdi Bahaghighat, *Majid Ghasemi*, and Figen Ozen.

Honors and scholarships	AIES conference Student Program winner	2026
	GSRDA Award x2 (University of Waterloo)	2026
	Int'l Doctoral Student Award (University of Waterloo)	2025 – 2029
	Winner of the Academic Excellence Gold Medal (WLU)	2025
	Graduate Scholarship (WLU)	2023 – 2025
	Graduate Fellowship (WLU)	2023 – 2025
	Differential Tuition Fee Waiver (WLU)	2023 – 2025
	Excellence Students Tuition Fee Waiver (IKIU)	2017–2022
Research experience	UWECML/CompThink lab	
	Supervisor: Prof. Mark Crowley (University of Waterloo) Sep 2025 – Present Developing socially informed and ethically grounded approaches to multi-agent reinforcement learning.	
	Edge Computing lab	
	Supervisor: Dr. Dariush Ebrahimi (WLU)	Sep 2023 – May 2025
	Conducted research on reinforcement learning for intelligent transportation systems, focusing on EV fleet management and routing optimization.	
Teaching experience	AIST lab	
	Supervisor: Dr. Mahdi Bahaghighat (IKIU)	Sep Jan. 2021 – March 2022
	Researched on phishing website detection with machine learning.	
	Teaching Assistant, IKIU, WLU, and UW	
	2023-Present	
Industry experience	Courses including AI, Data Mining, Algorithm Design and Analysis, and Discrete Mathematics	
	Kisoji Biotechnology Inc. , Machine Learning Department Waterloo, ON, Canada	
	AI Research Intern	May 2026 – Present
	Currently building the Expressibility Engine — a 3-module system for antibody expression optimization: domain-adaptive prediction with Bayesian project embeddings, RL-guided experimental design (GRPO + DPP diversity), and sparse graph-based causal driver discovery. Reduces wet-lab rounds via active learning while providing interpretable guidance for antibody engineers.	
	(R) Stands for Reviewer, (PC) stands for Program Committee	
Service to the Community	(PC) AAAI/ACM Conference on AI, Ethics, and Philosophy (AIES) 2026	

(R) Reinforcement Learning in Big Worlds at Reinforcement Learning Conference (RLC) 2026
(R) PhilML Workshop at ICML 2026
(R) Pluralistic Alignment Workshop at ICML 2026
(R) Post-AGI Science and Society (P-AGI) Workshop at ICLR 2026
(R) IEEE Canadian Journal of Electrical & Computer Engineering (CJECE)
(R) ACM Transactions on Intelligent Systems and Technology

Grants

HU University of Applied Sciences Utrecht Aug. 2022 – Aug. 2023

Recipient of a grant to conduct research on Woman, Life, Freedom movement in Iran with computational social science and machine learning approaches.